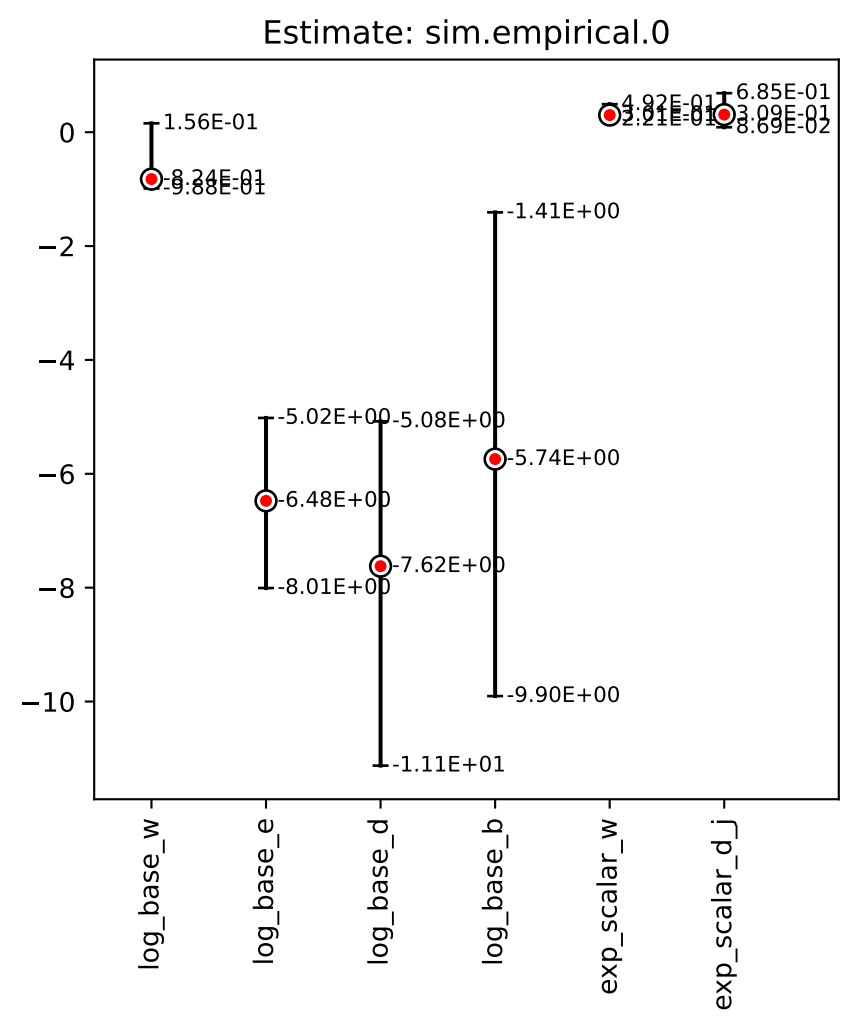
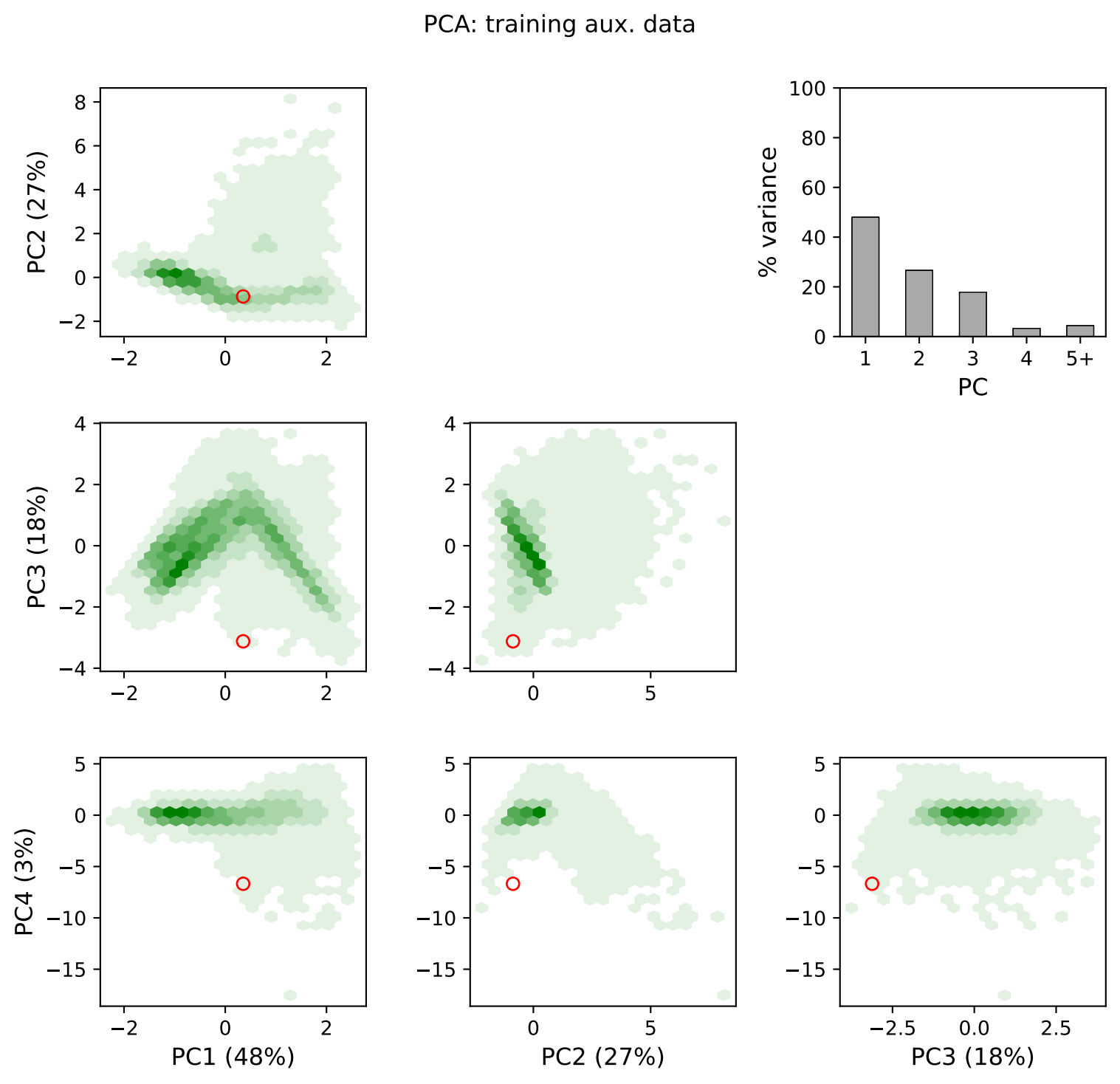
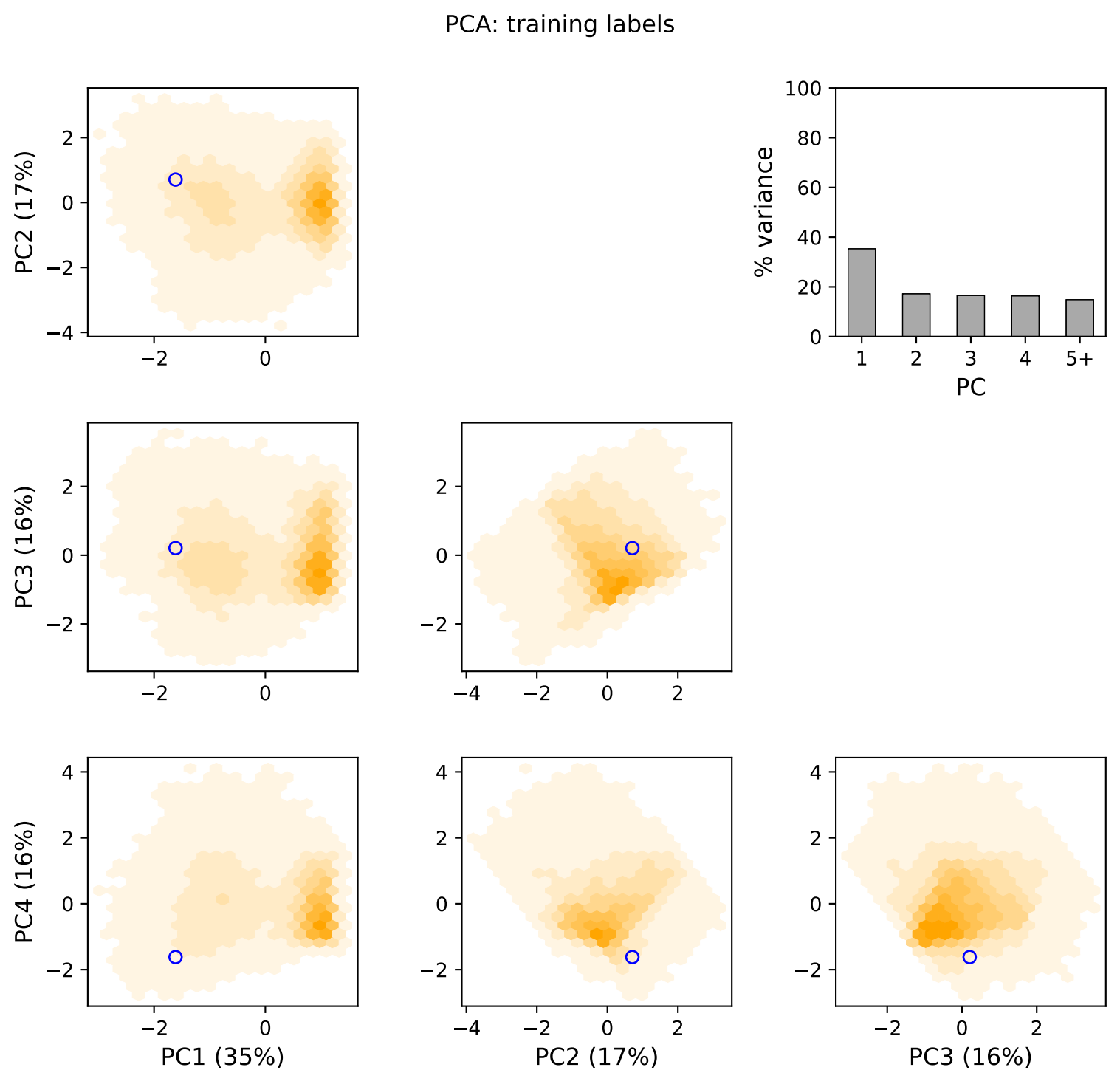
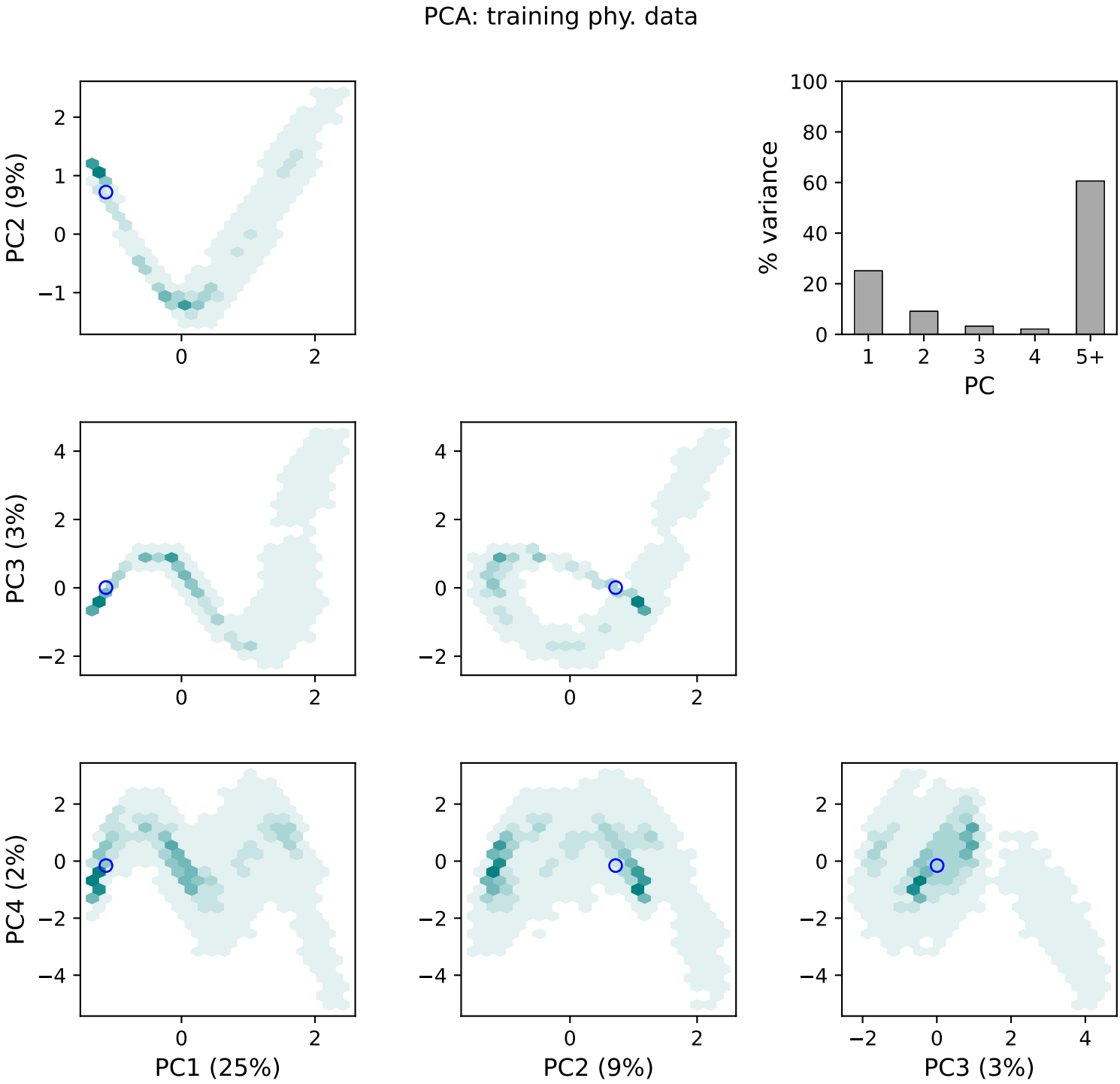
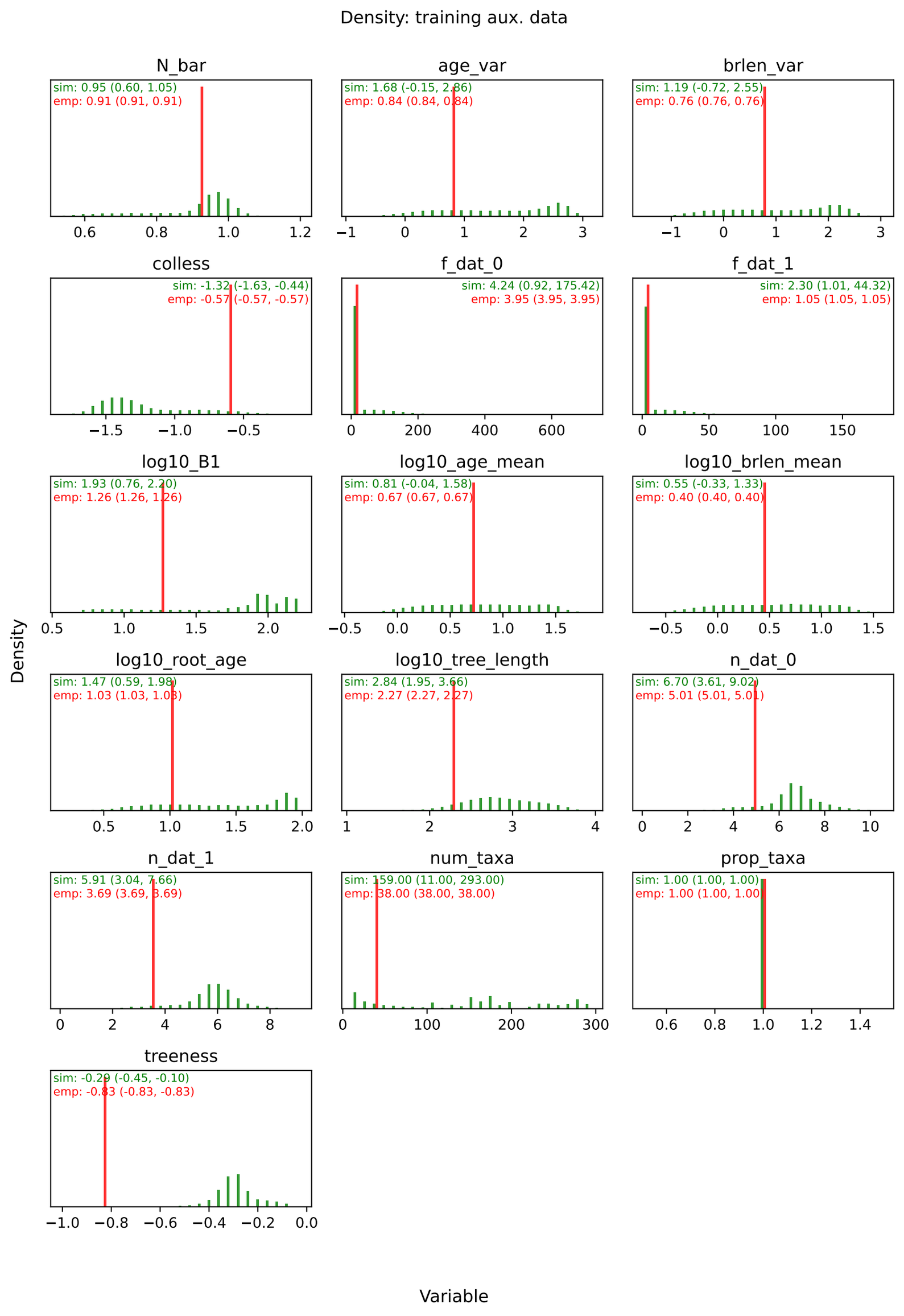
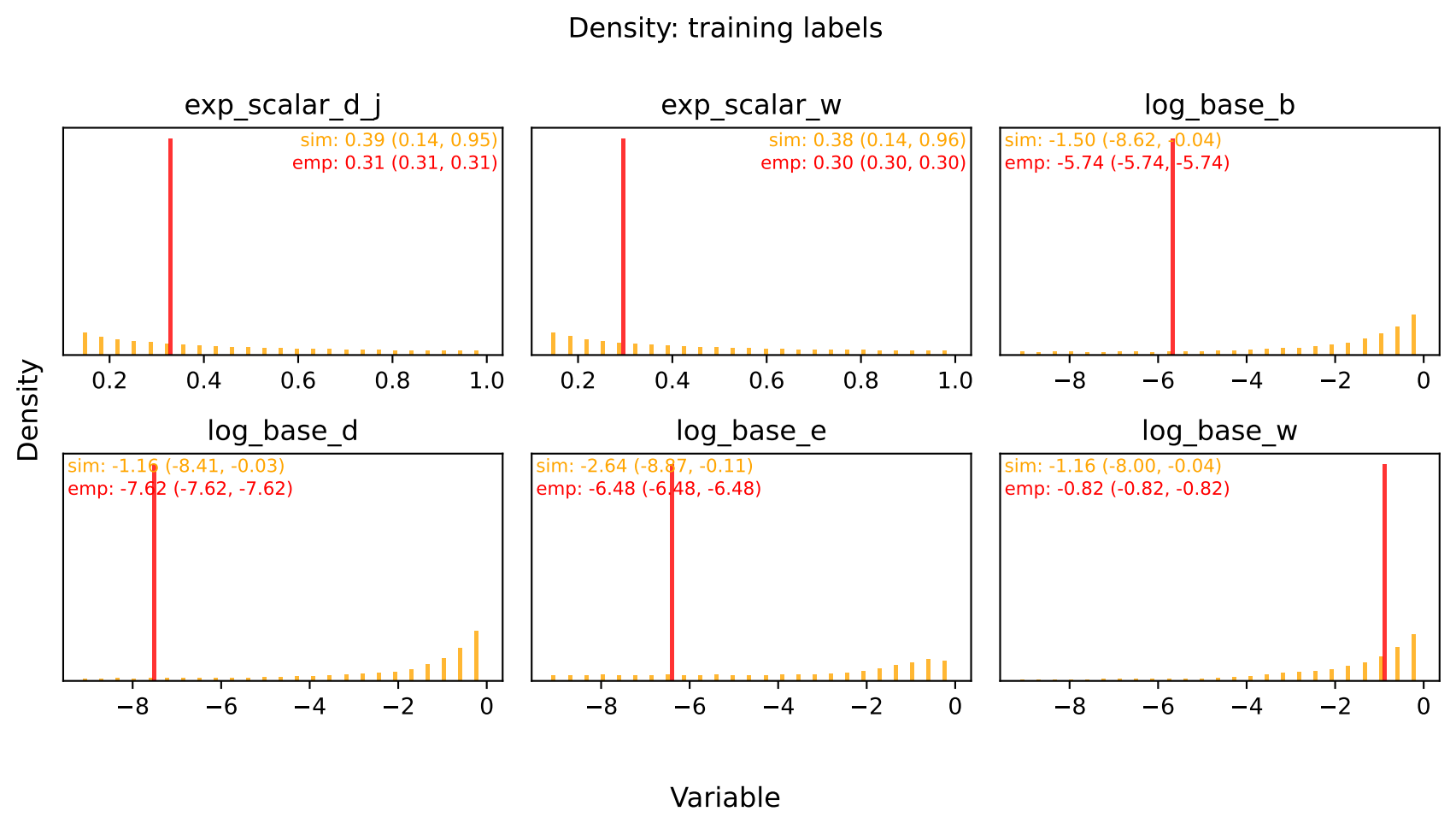

Estimate: sim.empirical.0
0
−2
−4
−6
−8
−10
1.56E-01
-8.24E-01
-9.68E-01
-5.02E+00
-6.48E+00
-8.01E+00
-5.08E+00
-7.62E+00
-1.11E+01
-1.41E+00
-5.74E+00
-9.90E+00
-4.92E-01
6.85E-01
3.09E-01
8.69E-02
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_w
exp_scalar_d_j

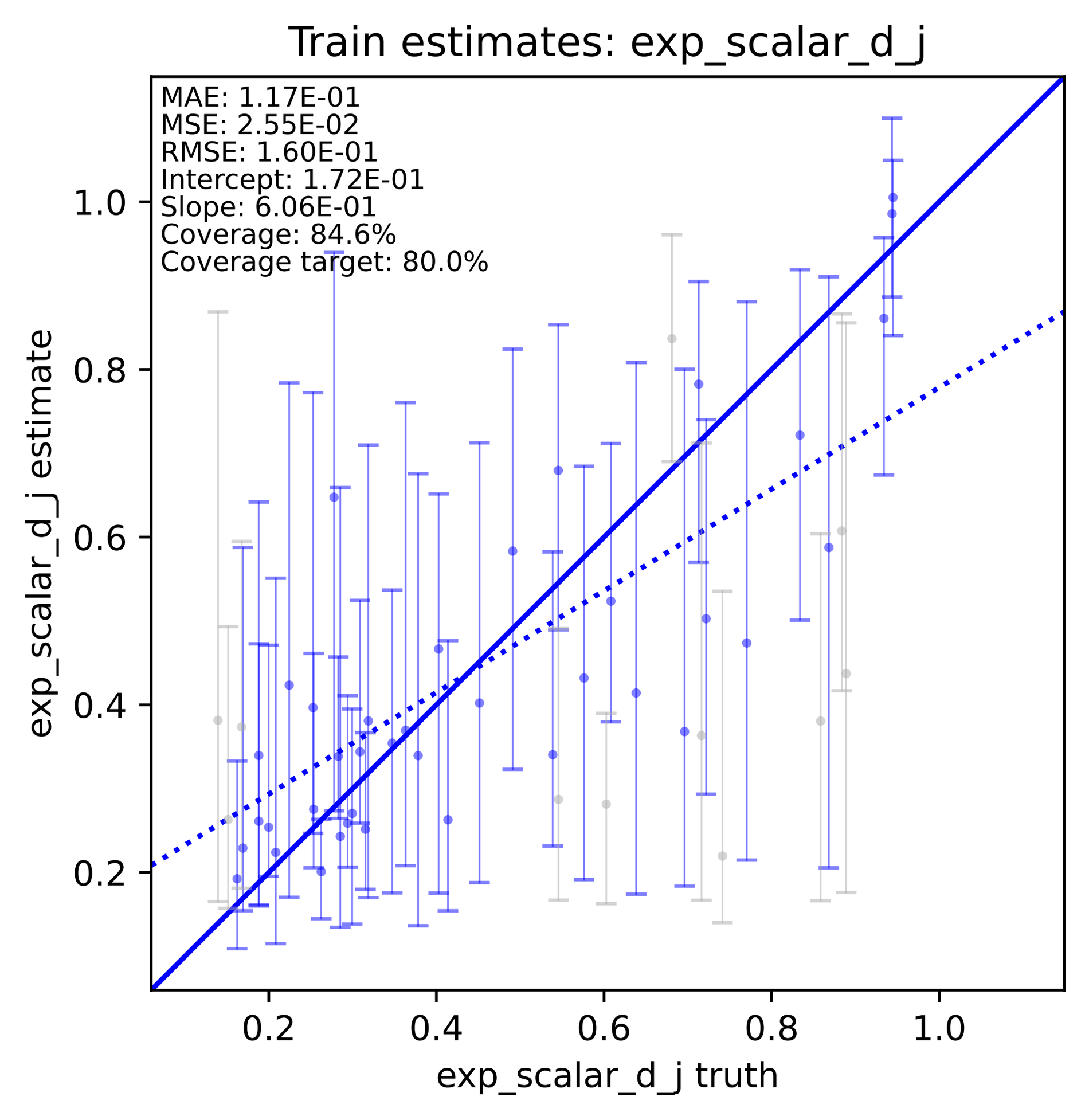

PCA: training aux. data
PC2 (27%)
% variance
PC
PC3 (18%)
PC4 (3%)
PC1 (48%)
PC2 (27%)
PC3 (18%)

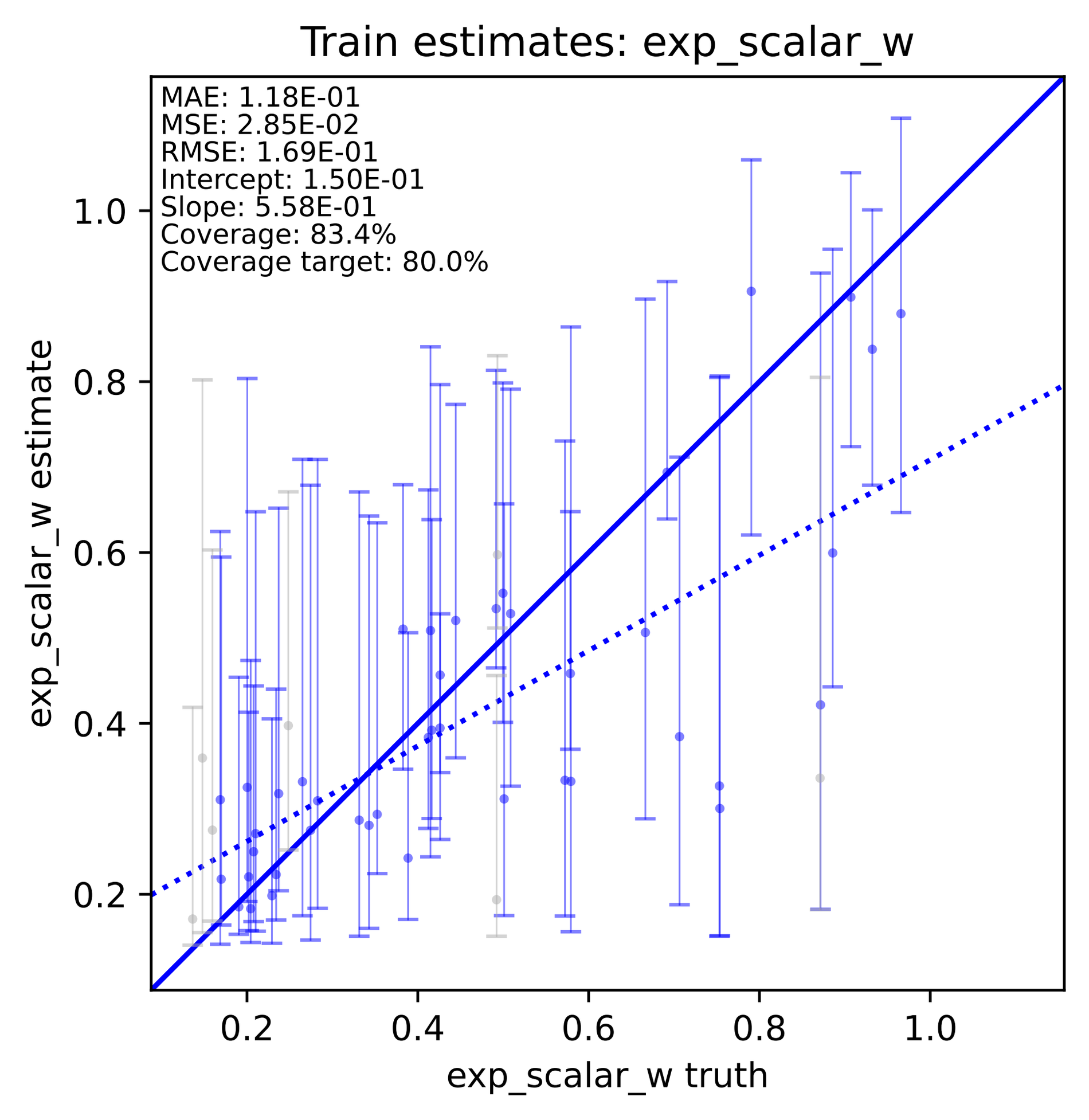

PCA: training labels
PC2 (17%)
% variance
PC
PC3 (16%)
PC4 (16%)
PC1 (35%)
PC2 (17%)
PC3 (16%)

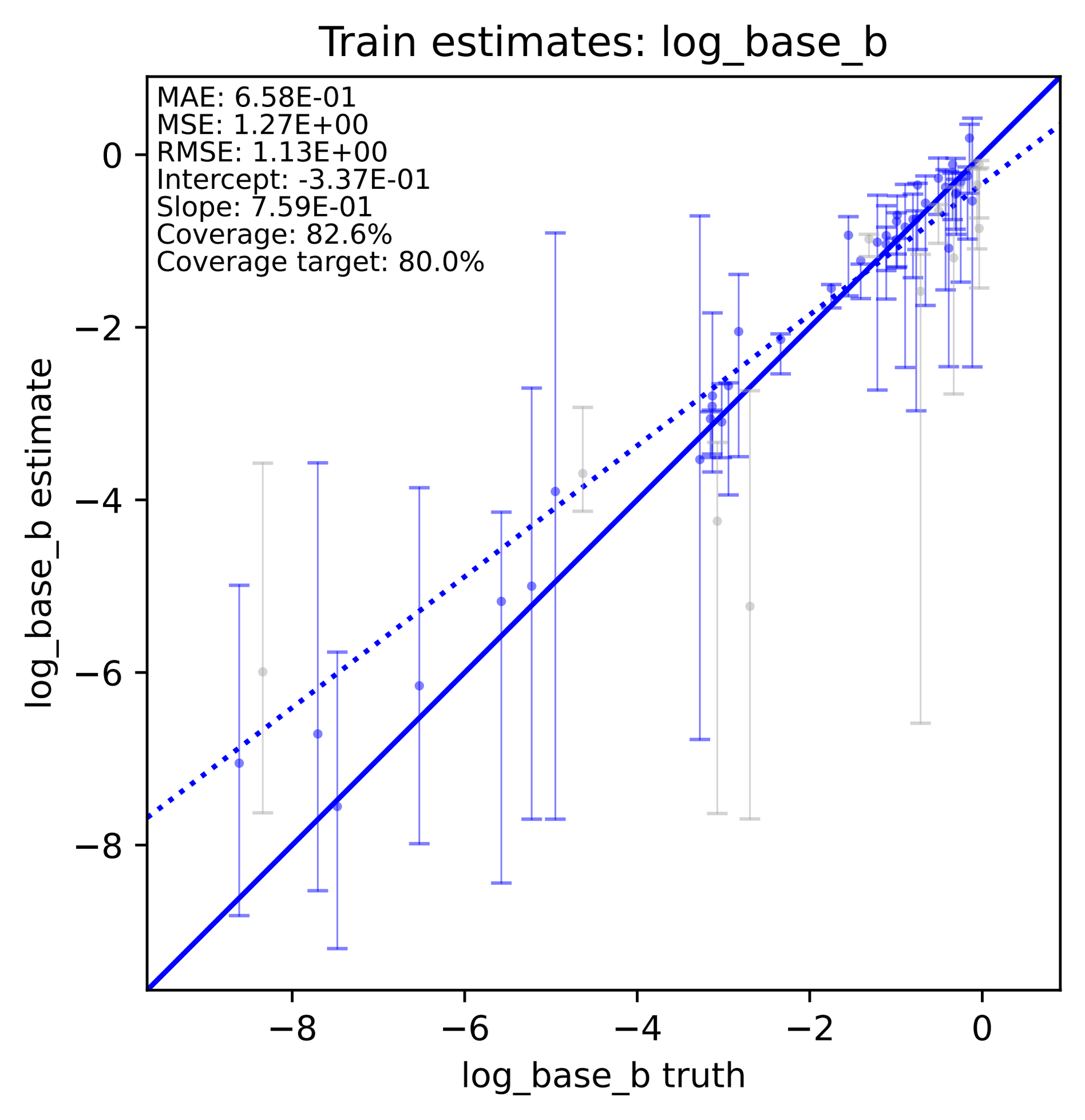

PCA: training phy. data
PC2 (9%)
% variance
PC
PC3 (3%)
PC4 (2%)
PC1 (25%)
PC2 (9%)
PC3 (3%)

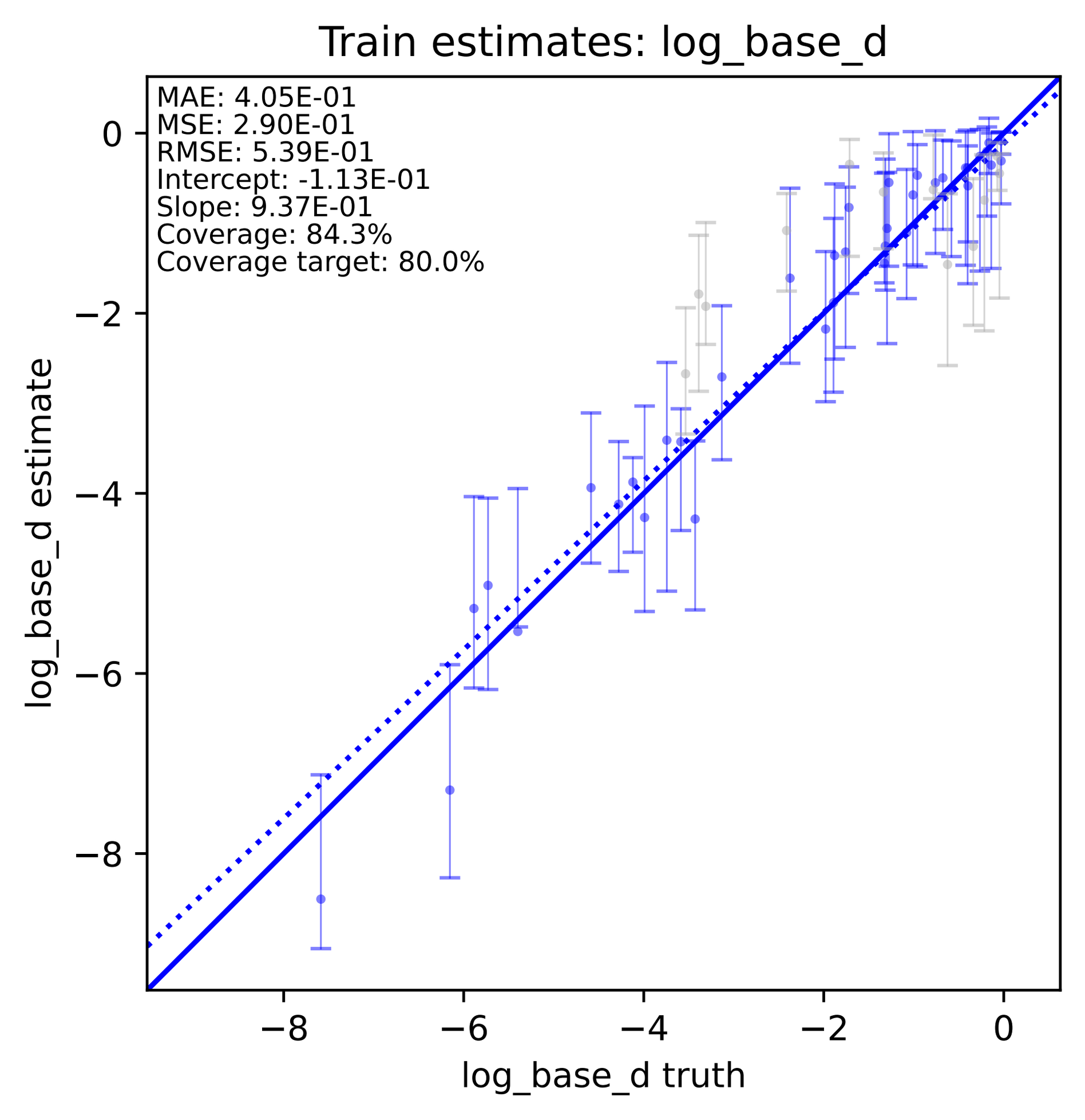

Density: training aux. data
N_bar
sim: 0.95 (0.60, 1.05)
emp: 0.91 (0.91, 0.91)
age_var
sim: 1.68 (-0.15, 2.86)
emp: 0.84 (0.84, 0.84)
brlen_var
sim: 1.19 (-0.72, 2.55)
emp: 0.76 (0.76, 0.76)
colless
sim: -1.32 (-1.63, -0.44)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 4.24 (0.92, 175.42)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.30 (1.01, 44.32)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 1.93 (0.76, 2.20)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.81 (-0.04, 1.58)
emp: 0.67 (0.67, 0.67)
log10_brlen_mean
sim: 0.55 (-0.33, 1.33)
emp: 0.40 (0.40, 0.40)
log10_root_age
sim: 1.47 (0.59, 1.98)
emp: 1.03 (1.03, 1.03)
log10_tree_length
sim: 2.84 (1.95, 3.66)
emp: 2.27 (2.27, 2.27)
n_dat_0
sim: 6.70 (3.61, 9.02)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 5.91 (3.04, 7.66)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 159.00 (11.00, 293.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.29 (-0.45, -0.10)
emp: -0.83 (-0.83, -0.83)
Density
Variable

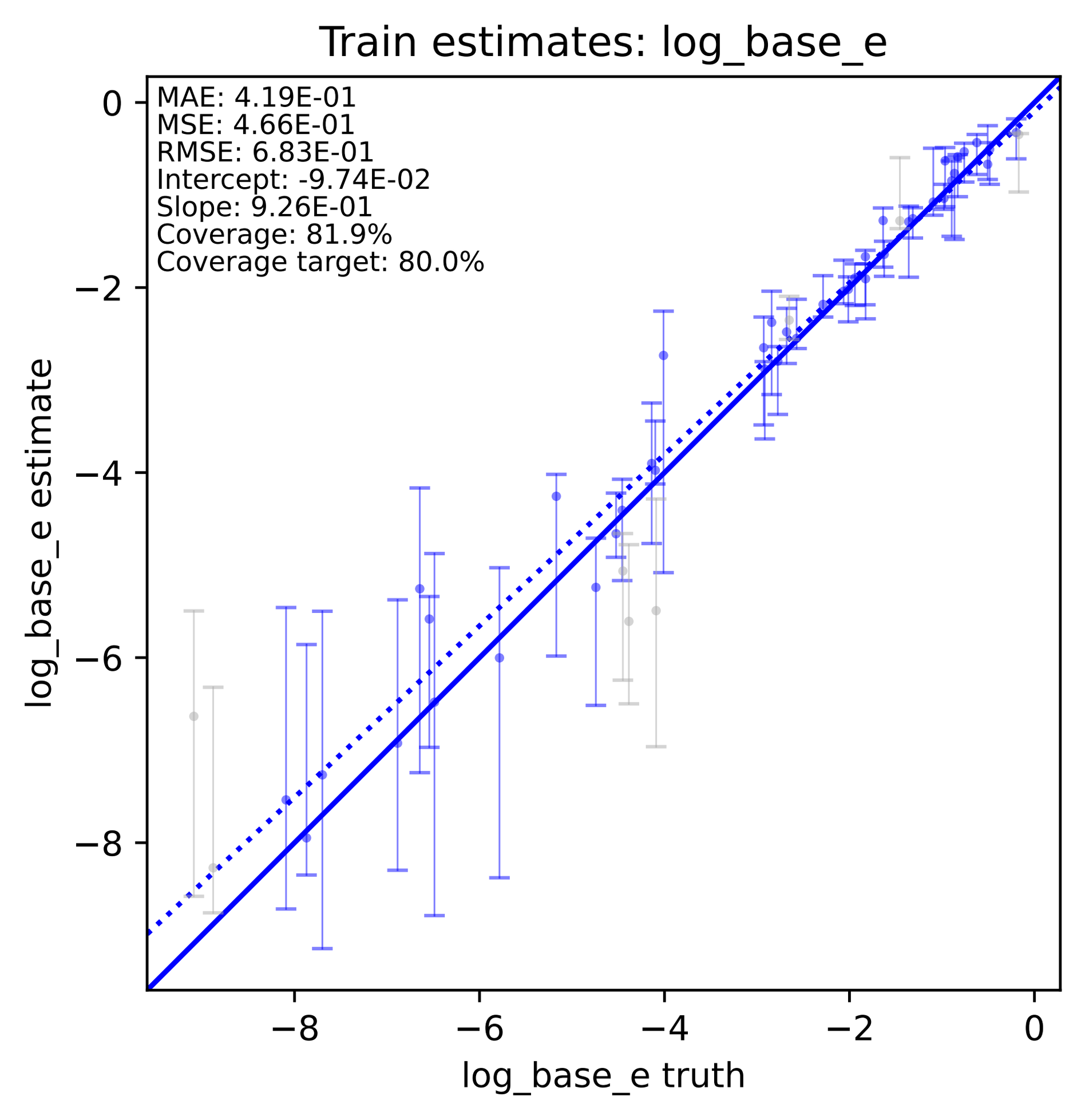

Density: training labels
exp_scalar_d_j
sim: 0.39 (0.14, 0.95)
emp: 0.31 (0.31, 0.31)
exp_scalar_w
sim: 0.38 (0.14, 0.96)
emp: 0.30 (0.30, 0.30)
log_base_b
sim: -1.50 (-8.62, 0.04)
emp: -5.74 (-5.74, -5.74)
log_base_d
sim: -1.16 (-8.41, -0.03)
emp: -7.62 (-7.62, -7.62)
log_base_e
sim: -2.64 (-8.87, -0.11)
emp: -6.48 (-6.48, -6.48)
log_base_w
sim: -1.16 (-8.00, -0.04)
emp: -0.82 (-0.82, -0.82)
Density
Variable

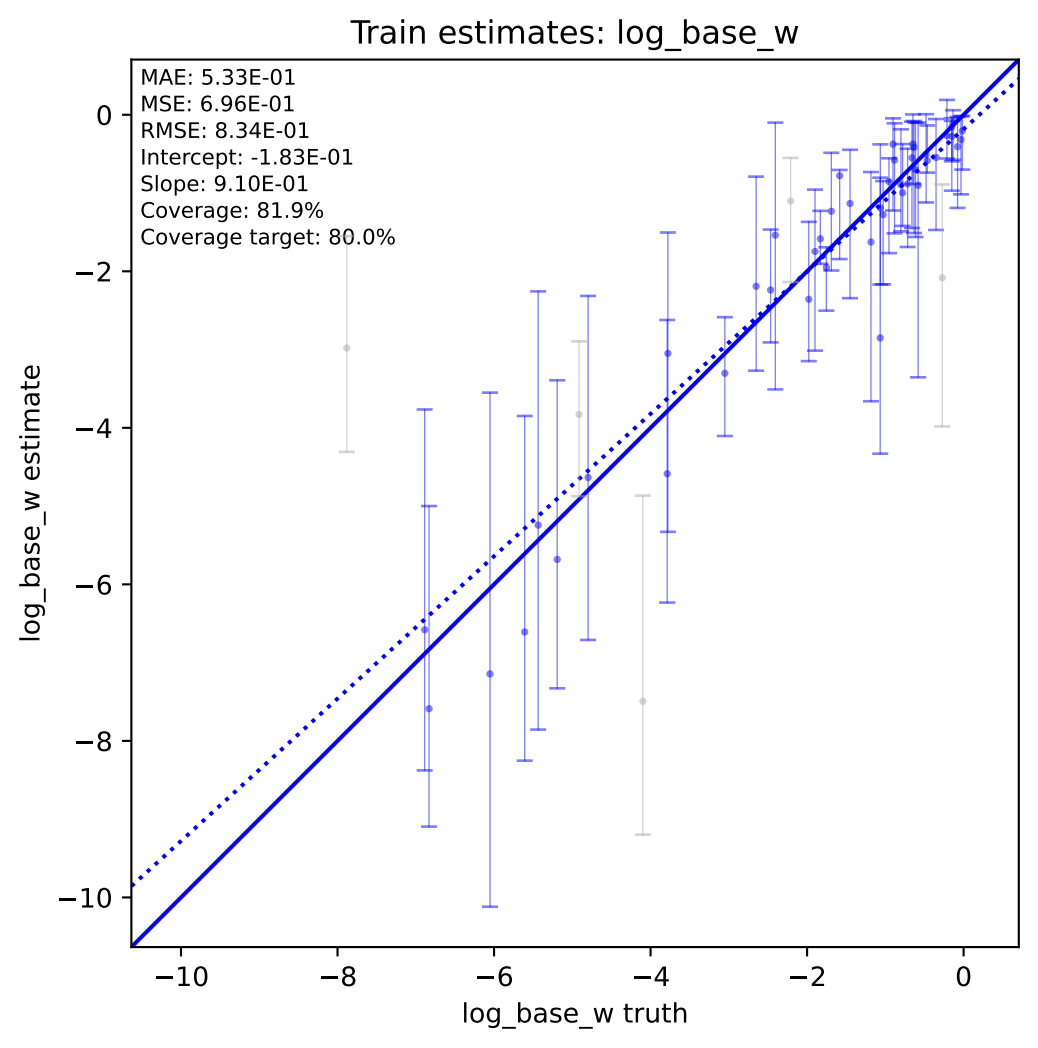

Train estimates: exp_scalar_d_j
MAE: 1.17E-01
MSE: 2.55E-02
RMSE: 1.60E-01
Intercept: 1.72E-01
Slope: 6.06E-01
Coverage: 84.6%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

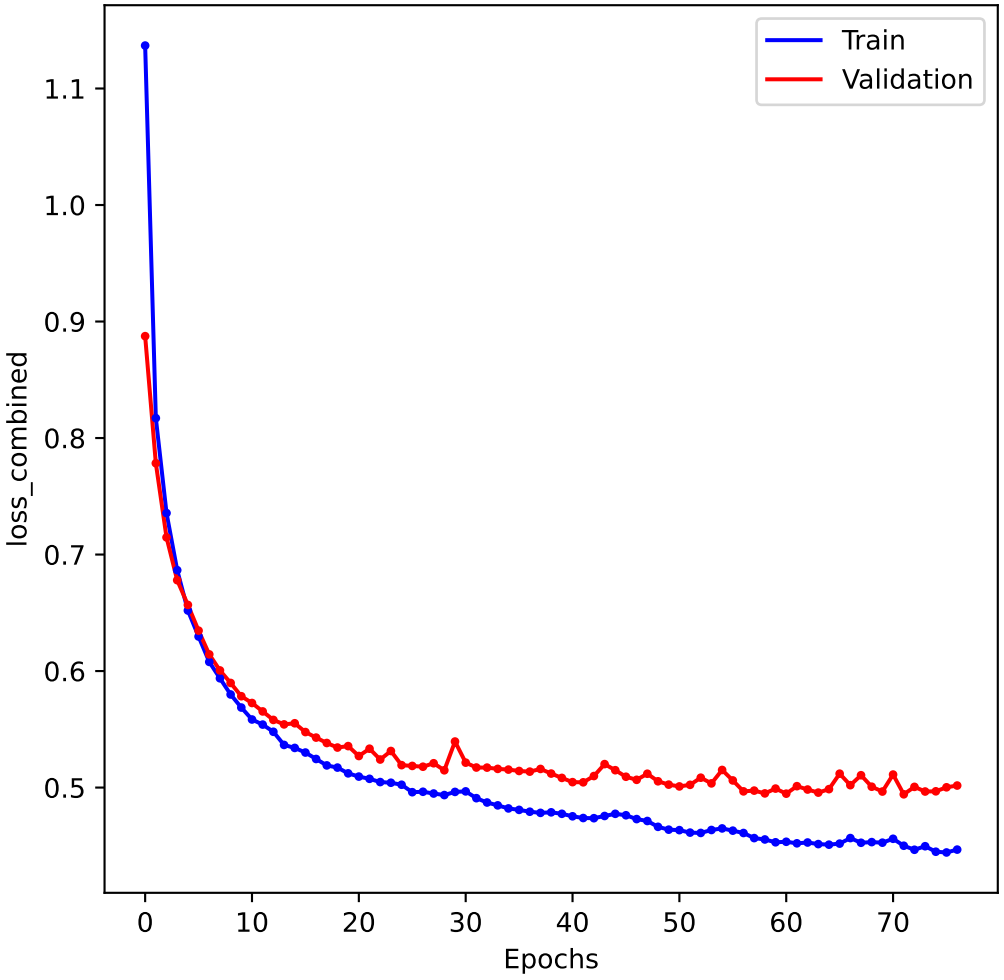

Train estimates: exp_scalar_w
MAE: 1.18E-01
MSE: 2.85E-02
RMSE: 1.69E-01
Intercept: 1.50E-01
Slope: 5.58E-01
Coverage: 83.4%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

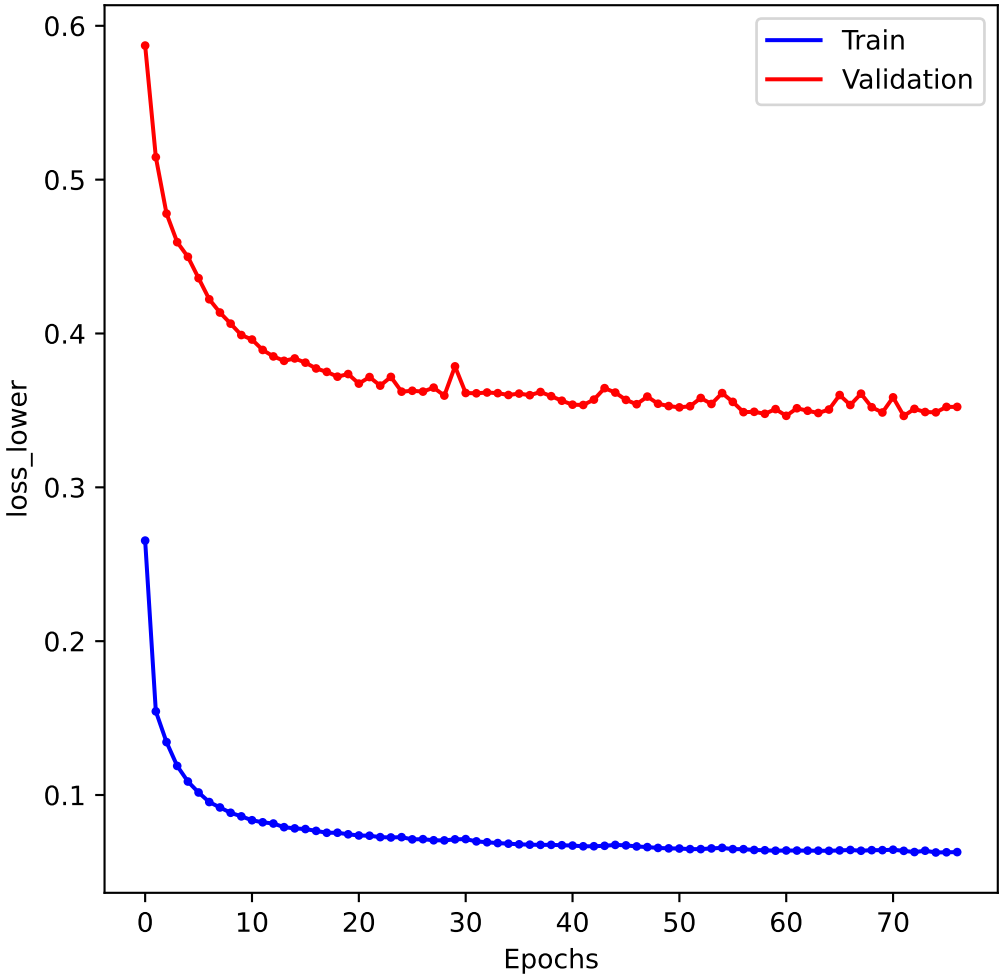

Train estimates: log_base_b
MAE: 6.58E-01
MSE: 1.27E+00
RMSE: 1.13E+00
Intercept: -3.37E-01
Slope: 7.59E-01
Coverage: 82.6%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

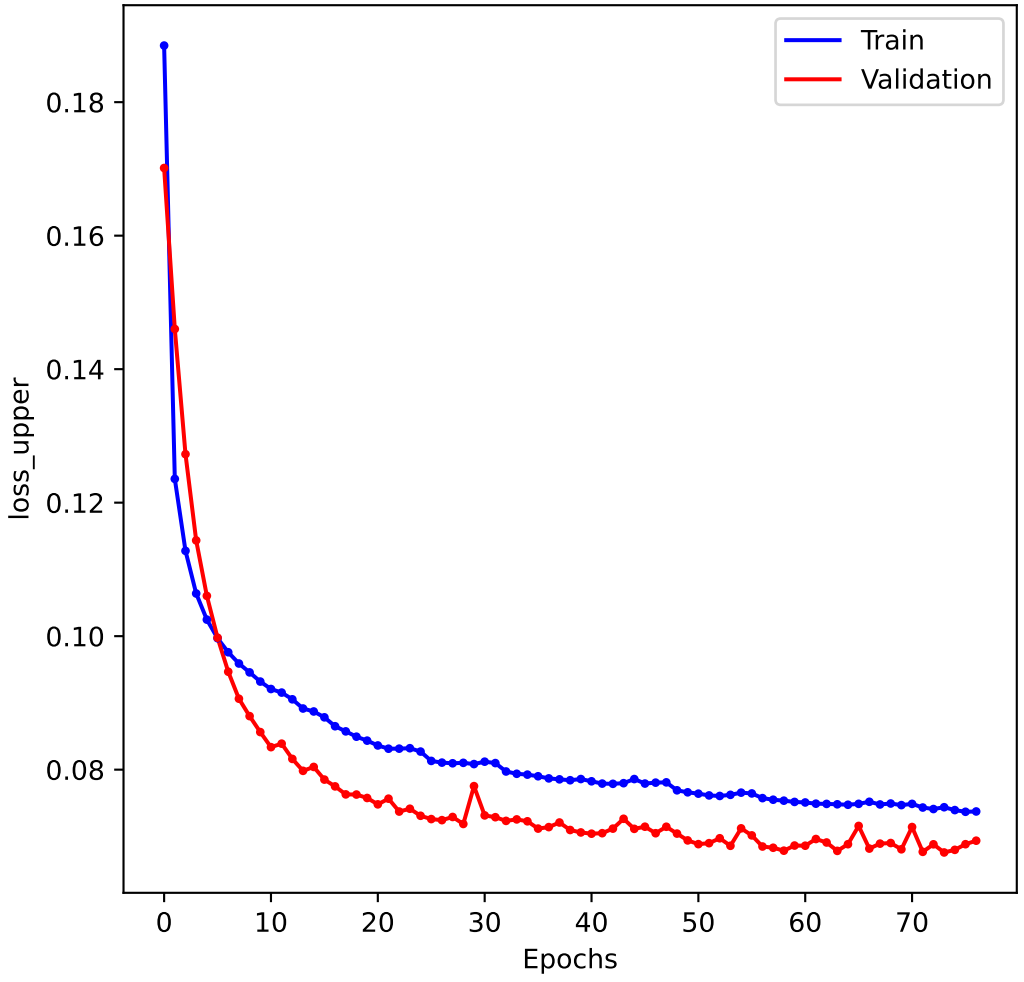

Train estimates: log_base_d
MAE: 4.05E-01
MSE: 2.90E-01
RMSE: 5.39E-01
Intercept: -1.13E-01
Slope: 9.37E-01
Coverage: 84.3%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

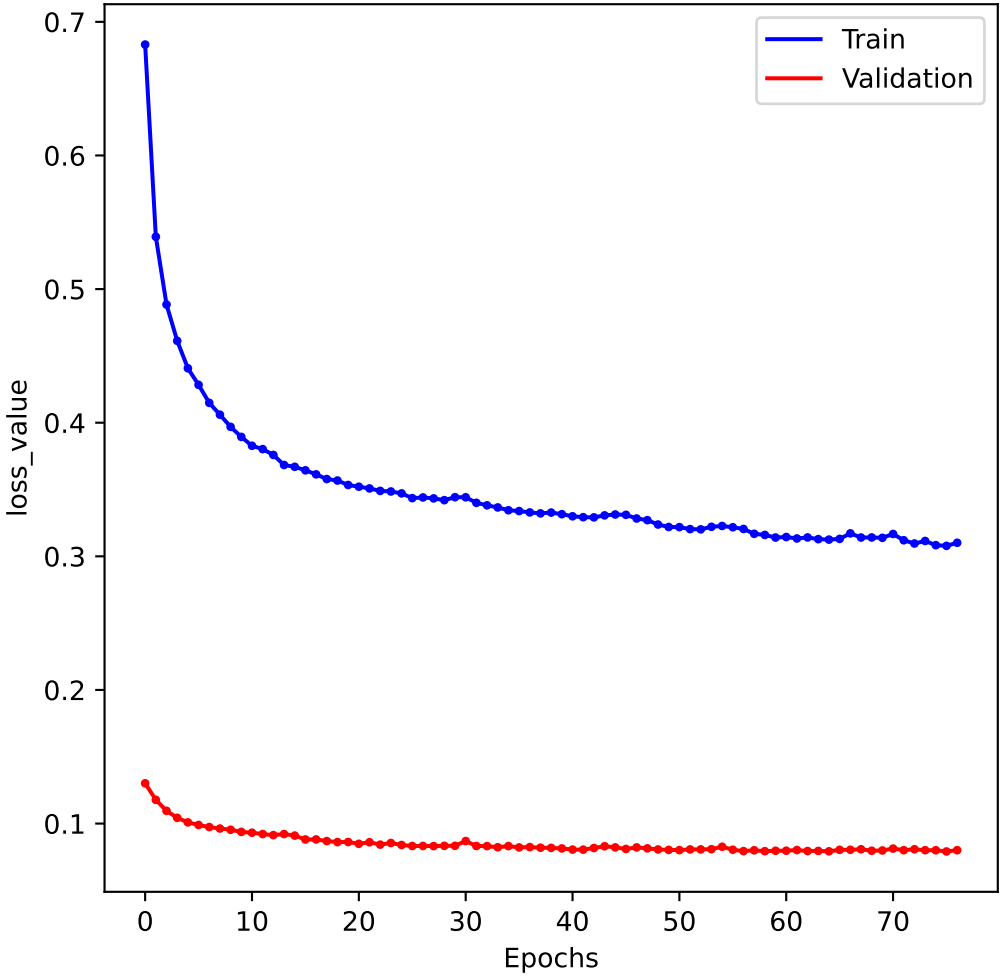

Train estimates: log_base_e
MAE: 4.19E-01
MSE: 4.66E-01
RMSE: 6.83E-01
Intercept: -9.74E-02
Slope: 9.26E-01
Coverage: 81.9%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

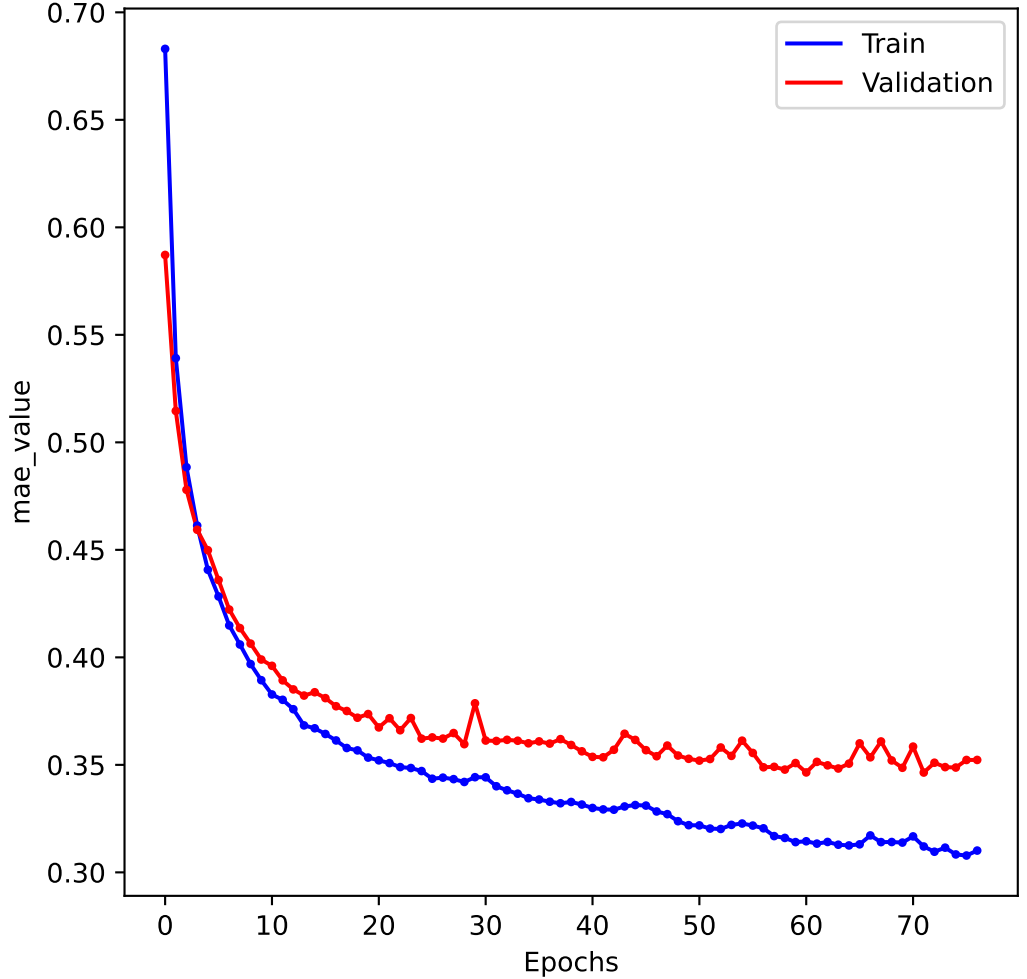

Train estimates: log_base_w
MAE: 5.33E-01
MSE: 6.96E-01
RMSE: 8.34E-01
Intercept: -1.83E-01
Slope: 9.10E-01
Coverage: 81.9%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

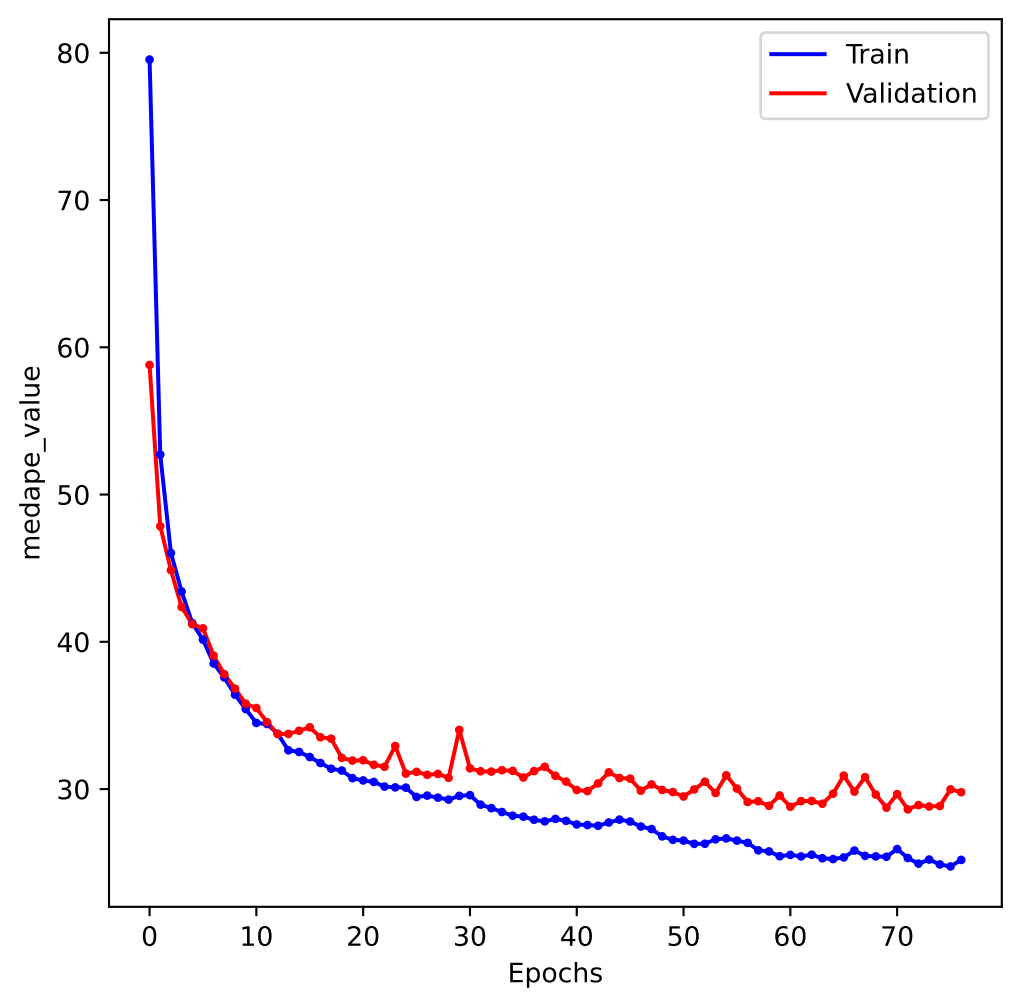

Train
Validation
loss_combined
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0
10
20
30
40
50
60
70
Epochs

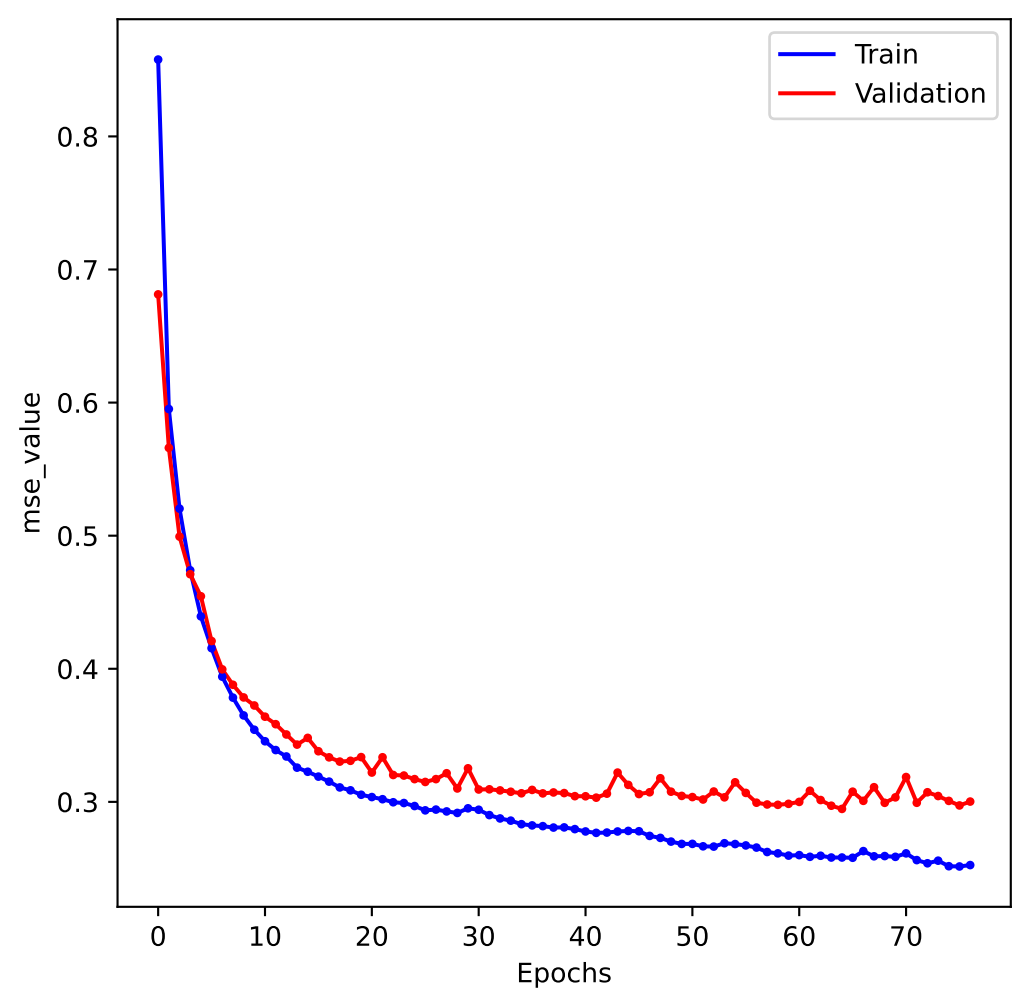

Train
Validation
loss_lower
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
70
Epochs

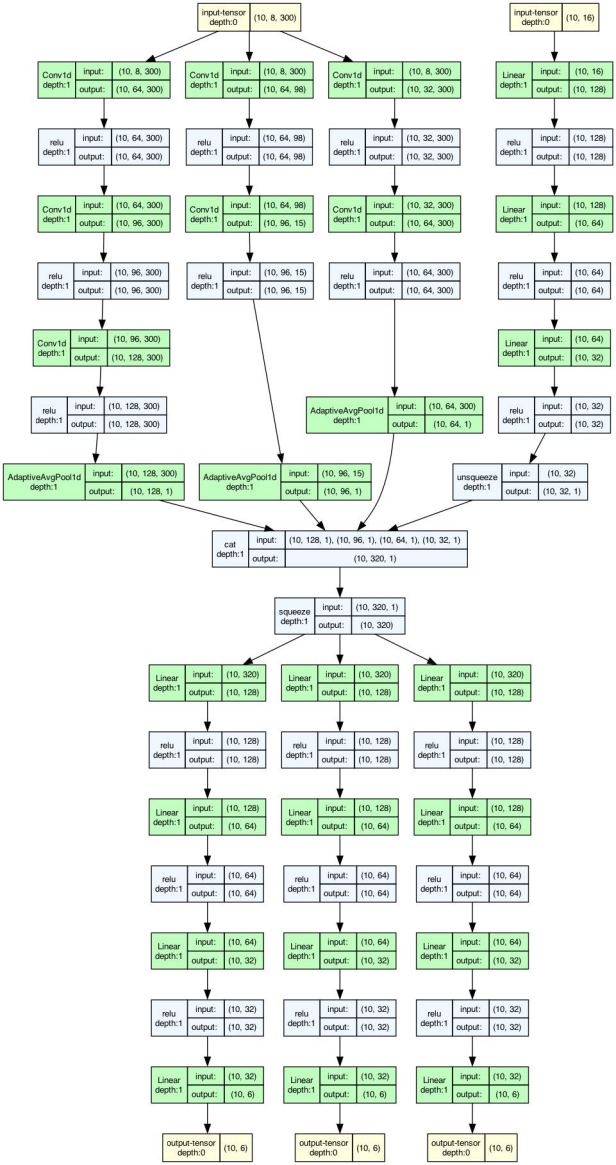

Train
Validation
loss_upper
0.18
0.16
0.14
0.12
0.10
0.08
0
10
20
30
40
50
60
70
Epochs


Train
Validation
loss_value
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
60
70
Epochs


Train
Validation
mae_value
Epochs
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0
10
20
30
40
50
60
70


Train
Validation
medape_value
Epochs
80
70
60
50
40
30
0
10
20
30
40
50
60
70


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0.3
0
10
20
30
40
50
60
70
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:1
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)